A Test of exjobb_eit.typ, a Typst Template made by Lucas Ekholm

Student A
Student.A@mail.com

Student B
Lund University
Student.B@mail.com

Department of Electrical and Information Technology
Lund University

Some company?

Academic Supervisor: Some Supervisor

Company Supervisor: Another Supervisor

Examinor: Some examinor

October 22, 2022

This thesis is typeset using Typst 0.14.0

© 2022
Printed in Sweden
Tryckeriet i E-huset, Lund

Frontmatter

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

AC Actually Confusing

CODE Computers Only Do Exactly (what you tell them)

EIT Engineering Is (never) Trivial

EMC Engineers' Major Complaint

IC Infinite Confusion

LED Let's Emit Dazzle

MOSFET Mostly Our Small Friend Enabling Transistor

PING Packet Is Now Going

> "To mature as an electrical engineer is to realize everything is about impedances."
>
> — Markus Törmänen[1], spring 2025

[1] Translated. Original quote in Swedish.

Table of Contents

1 Introduction **1**
1.1 Test section 1
1.2 Another test section 2
1.3 Styling parameters 3

2 A New Chapter with a Short Title **5**
2.1 New section 5

3 A word on Typst referencing **7**

References **9**

A Backmatter **11**

List of Figures

1.1 An example figure using the package Lilaq. 2
1.2 A shorter caption, but links to the same figure! 2

2.1 A figure together with another figure. 5
2.2 This figure can have a long caption and still fit. 5

A.1 This is the last figure. 11

List of Tables

1.1 Document types, their size and color. 3
1.2 Fonts used. 3

2.1 A simple table. 5

Chapter 1

Introduction

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et.

$$\operatorname{div} \boldsymbol{A} \coloneqq \lim_{\Delta v \to 0} \frac{\oiint_S \boldsymbol{A} \cdot d\boldsymbol{s}}{\Delta v} \tag{1.1}$$

$$\nabla \times \boldsymbol{B} = \mu_0 \left(\boldsymbol{J} + \varepsilon_0 \frac{\partial \boldsymbol{E}}{\partial t} \right) \tag{1.2}$$

The equations above are Equation (1.1) and Equation (1.2). This is Chapter 1 but below is Section 1.1 and below that is Section 1.1.1.

1.1 Test section

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

1.1.1 A section which won't appear in the outline

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

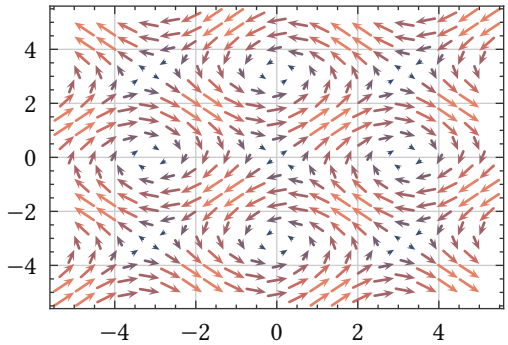

Figure 1.1: An example figure using the package Lilaq. [1]

1.2 Another test section

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et.[2]

Figure 1.2: A figure with a longer caption. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

[2] https://www.lth.se

1.3 Styling parameters

Table 1.1: Document types, their size and color.

	Headings	**Sub-headings**	**Sub-sub-headings**	**Body**	**Secondary text**	**Tertiary**
Size	`16pt` (Body), `42.52pt` (Chapter number)	`14pt`	`12pt`	`10pt`	`8pt`	-
Color						

Table 1.2: Fonts used.

	Body	**Secondary text**	**Chapter numbering**	**Math**	**Raw**
Font	EB Garamond	Lato	Libertinus Math	Libertinus Math	`Source Code Pro`
Usage	Main text	Headings, captions, header, footer	Chapter number	Equations, in-line math	Code blocks

Chapter 2

New Chapter with a Long Title that Spanns over More Than One Line

Figure 2.1: A figure together with another figure.

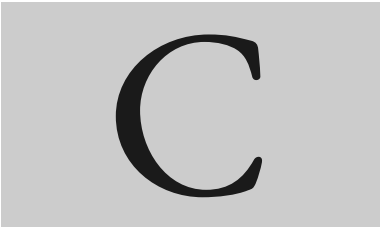

Figure 2.2: This figure can have a long caption and still fit if done this way. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do.

Above Figure 2.1 is next to Figure 2.2. By using the `#flexCaption(long caption, short caption)` function you can make flexible captions. You can also make longer/shorter headings! See Chapter 2.

2.1 New section

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et.

Table 2.1: A simple table.

t_0	t_1	t_2
y_0	y_1	y_2

Chapter 3

A word on Typst referencing

In References you can find links. [2] provides a guide for past LaTeX users on the differences between the two typesetting languages, including what's native or not native, which macros map to which function etc. [3] and [4] links to information on how to cite and reference different sources. Typst includes support for both its native YAML-based format and BibTeX. [5] gives a link to the Typst Universe. On this page you can find packages which aid in your workflow, both in terms of styling and automation. For example Figure 1.1 was made using lilaq, a package used to make graphs inside the Typst environment. Another good package is Zap which is used to make circuit diagrams much like TikZ. [6]

For documentation on all functions, markup- and styling commands, see [7].

References

[1] "Lilaq - Typst Universe." [Online]. Available: https://typst.app/universe/package/lilaq/

[2] "Guide for LaTeX users." [Online]. Available: https://typst.app/docs/guides/guide-for-latex-users/

[3] "Cite function." [Online]. Available: https://typst.app/docs/reference/model/cite/

[4] "Bibliography function." [Online]. Available: https://typst.app/docs/reference/model/bibliography/

[5] "Typst Universe." [Online]. Available: https://typst.app/universe/

[6] "Zap - Typst Universe." [Online]. Available: https://typst.app/universe/package/zap

[7] "Typst Documentation." [Online]. Available: https://typst.app/docs

Appendix A

Backmatter

```
/* Function to sort array using insertion sort */
void insertionSort(int arr[], int n)
{
    for (int i = 1; i < n; ++i) {
        int key = arr[i];
        int j = i - 1;

        /* Move elements of arr[0..i-1], that are
           greater than key, to one position ahead
            of their current position */
        while (j >= 0 && arr[j] > key) {
            arr[j + 1] = arr[j];
            j = j - 1;
        }
        arr[j + 1] = key;
    }
}
```

Listing A.1: Code snippet for insertionsort written in C.

Figure A.1: This is the last figure.